IME 775 — Assignment 3

Probability Distributions & Bayesian Tools (Chapters 5–6)

Course: IME 775 **Topics:** Probability Distributions, Bayes' Theorem, Entropy, Cross-Entropy, KL Divergence, MLE, MAP

Part A: Probability and Distributions

Problem 1. A medical imaging system classifies chest X-rays into three categories with the following joint probability table over diagnosis and patient age group:

	Young (A_1)	**Middle (A_2)**	**Senior (A_3)**
Normal (D_1)	0.25	0.15	0.05
Pneumonia (D_2)	0.05	0.10	0.15
Tumor (D_3)	0.02	0.08	0.15

(a) Compute the marginal distributions $P(D)$ and $P(A)$.

(b) Compute $P(D_2 \mid A_3)$ — the probability of pneumonia given a senior patient.

(c) Are diagnosis and age group independent? Justify with a specific check.

(d) If you observe a senior patient, which diagnosis is most likely?

Problem 2. Let $X \sim \mathcal{N}(8, 16)$.

(a) State μ and σ.

(b) Using the 68-95-99.7 rule, find $P(4 \leq X \leq 12)$ and $P(X > 16)$.

(c) A second variable $Y \sim \mathcal{N}(8, 16)$ is independent of X. Write the joint distribution of (X, Y) as a multivariate Gaussian, specifying $\vec{\mu}$ and Σ.

(d) Describe the shape of the equal-probability contours for this joint distribution.

Problem 3. Consider a 2D Gaussian with $\vec{\mu} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $\Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}$.

(a) Compute Σ^{-1}.

(b) Compute the Mahalanobis distance of the point $\vec{x} = (3, 5)^T$ from the mean.

(c) Find the eigenvalues of Σ. What are the semi-axis lengths of the 1-σ ellipse?

(d) Is the ellipse axis-aligned or rotated? Explain.

Problem 4. A sentiment classifier outputs probabilities $\vec{\theta} = (0.65, 0.25, 0.10)$ for classes (positive, neutral, negative).

(a) Write this as a Categorical distribution.

(b) In a batch of 200 reviews, what is the expected count and standard deviation for each class? (Use the Multinomial distribution.)

(c) If the true label is "positive," write the one-hot vector $\vec{y}$ and compute the cross-entropy loss $\mathcal{L} = -\sum_k y_k \log \hat{y}_k$.

Problem 5. A coin is flipped 20 times, resulting in 13 heads and 7 tails.

(a) Model each flip as $X_i \sim \mathrm{Ber}(\theta)$. What is the MLE estimate $\hat{\theta}_{MLE}$?

(b) Compute $E[X]$ and $\mathrm{Var}(X)$ using the MLE estimate.

(c) Using the Binomial distribution with $n = 20$ and $\hat{\theta}_{MLE}$, what is $P(\text{exactly 10 heads})$? (Set up the expression; you may leave the numerical answer in factorial/exponential form.)

Part B: Bayes' Theorem and Entropy

Problem 6. A factory has two machines. Machine A produces 60% of all items and Machine B produces 40%. Machine A has a 3% defect rate and Machine B has a 5% defect rate.

(a) What is the probability that a randomly selected item is defective?

(b) Given that an item is defective, what is the probability it came from Machine B? Use Bayes' theorem and clearly label the prior, likelihood, and posterior.

(c) If 10 defective items are found, and 7 came from Machine B, update your belief using the observed data. How does this compare to your answer in (b)?

Problem 7. A discrete random variable X has the following distribution:

x	**A**	**B**	**C**	**D**
$P(X = x)$	0.4	0.3	0.2	0.1

(a) Compute the Shannon entropy $H(X)$ in bits (use $\log_2$).

(b) What would the entropy be if X were uniformly distributed over $\{A, B, C, D\}$? Why is this value higher?

(c) A sensor reading is modeled as $Y \sim \mathcal{N}(0, 9)$. Compute the differential entropy $H(Y)$ in nats.

(d) If the variance increases to $\sigma^2 = 25$, how does the entropy change? Explain intuitively.

Problem 8. A 3-class classifier produces softmax output $\hat{\vec{y}} = (0.7, 0.2, 0.1)$ for an input whose true label is class 1.

(a) Write the one-hot label vector $\vec{y}$.

(b) Compute the cross-entropy loss $H(\vec{y}, \hat{\vec{y}}) = -\sum_k y_k \log \hat{y}_k$ (use natural log).

(c) If the model instead output $\hat{\vec{y}} = (0.9, 0.05, 0.05)$, recompute the cross-entropy loss. Why is it lower?

(d) Show that minimizing cross-entropy loss is equivalent to maximizing the log-likelihood of the correct class.

Part C: KL Divergence, MLE, and MAP

Problem 9. Consider two discrete distributions over $\{A, B, C\}$:

- $p = (0.5, 0.3, 0.2)$ (true distribution)
- $q = (0.4, 0.4, 0.2)$ (model distribution)

(a) Compute $D_{KL}(p\|q)$ in nats.

(b) Compute $D_{KL}(q\|p)$ in nats.

(c) Verify that $D_{KL}(p\|q) \neq D_{KL}(q\|p)$. What does this asymmetry mean practically?

(d) Compute the cross-entropy $H(p, q)$ and verify the identity $D_{KL}(p\|q) = H(p, q) - H(p)$.

Problem 10. You observe 5 data points drawn from a Gaussian distribution: $\{3.2, 4.8, 5.1, 3.9, 4.5\}$.

(a) Compute the MLE estimates $\hat{\mu}_{MLE}$ and $\hat{\sigma}^2_{MLE}$.

(b) Write the log-likelihood function $\ell(\mu, \sigma^2)$ for a Gaussian model. Show why maximizing it yields the sample mean.

(c) Why is $\hat{\sigma}^2_{MLE}$ a biased estimator? What is the unbiased version?

(d) Explain the connection: "Minimizing MSE loss in regression is equivalent to MLE under a Gaussian noise assumption."

Problem 11. A coin is flipped 10 times, yielding 7 heads. You use a Beta prior $\mathrm{Beta}(\alpha, \beta)$ on the bias θ.

(a) With a uniform prior ($\alpha = 1, \beta = 1$), compute the MAP estimate $\hat{\theta}_{MAP}$.

(b) With a prior biased toward a fair coin ($\alpha = 5, \beta = 5$), compute $\hat{\theta}_{MAP}$.

(c) Compare both MAP estimates with the MLE estimate $\hat{\theta}_{MLE} = 0.7$. Explain the shrinkage effect of the prior.

(d) Explain the analogy: "Gaussian prior on neural network weights ↔ L2 regularization."

Problem 12. Two Gaussians: $p = \mathcal{N}(0, 1)$ and $q = \mathcal{N}(1, 4)$.

(a) Compute $D_{KL}(p \| q)$ using the closed-form formula:

$$D_{KL}(p \| q) = \log \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$$

(b) Compute $D_{KL}(q \| p)$.

(c) Which direction has larger KL divergence? Give an intuitive explanation.

(d) In a VAE, the encoder produces $q = \mathcal{N}(\mu, \sigma^2)$ and we regularize toward $p = \mathcal{N}(0, 1)$. Write the KL regularization term and explain its role.

Part D: Connections and Conceptual Questions

Problem 13. For each ML scenario, identify which loss function / estimation method is being used and explain the probabilistic justification:

(a) Training a logistic regression model with binary cross-entropy loss.

(b) Training a neural network with MSE loss and weight decay ($\lambda \|\vec{w}\|^2$).

(c) A language model that predicts the next word from a vocabulary of 50,000 words.

(d) A VAE whose loss has a reconstruction term plus a KL divergence term.

Problem 14. True or False (with justification):

(a) KL divergence is a true distance metric (i.e., it satisfies symmetry and triangle inequality).

(b) The MAP estimate with a uniform prior is identical to the MLE estimate.

(c) Cross-entropy is always greater than or equal to entropy: $H(p, q) \geq H(p)$.

(d) If two Gaussians have the same mean but different variances, their KL divergence is zero.

(e) Mutual information $I(X; Y)$ can be negative.

(f) The MLE estimate of the variance of a Gaussian divides by N rather than $N - 1$, making it biased.

Part E: PyTorch Implementation

Problem 15. Write PyTorch code to:

(a) Generate 1000 samples from $\mathcal{N}(5, 4)$ and compute the MLE estimates of μ and σ^2. Compare with the true values.

(b) Given true distribution $p = [0.25, 0.25, 0.25, 0.25]$ and model distribution $q = [0.1, 0.2, 0.3, 0.4]$, compute the cross-entropy $H(p, q)$, entropy $H(p)$, and KL divergence $D_{KL}(p\|q)$. Verify $D_{KL} = H(p, q) - H(p)$.

(c) Implement a function that computes the MAP estimate for a Bernoulli parameter with a Beta prior, given observed data. Test with $k = 7$ heads out of $n = 10$ flips, and priors $\mathrm{Beta}(1, 1)$, $\mathrm{Beta}(5, 5)$, and $\mathrm{Beta}(10, 2)$.
